
 FOUNDATIONS — **DETAILED & EXPANDED**

Category	**Library / Tool**	**Topic**	➢ **Sub-Topic / Depth (What you MUST know)**
Programming	Python	Syntax	➢ variables, data types, indentation rules
Programming	Python	Control Flow	➢ if–else, for, while, break, continue, pass
Programming	Python	Iteration	➢ range(), iterables, iterators
Programming	Python	Functions	➢ function definition, return values
Programming	Python	Function Arguments	➢ positional, keyword, default, *args, **kwargs
Programming	Python	Lambda Functions	➢ inline functions, use cases
Programming	Python	Scope	➢ local, global, nonlocal
Programming	Python	Built-in Functions	➢ map, filter, zip, enumerate
Programming	Python	Data Structures	➢ list, tuple, set, dict
Programming	Python	List Comprehension	➢ nested comprehensions
Programming	Python	String Handling	➢ slicing, formatting, f-strings
Programming	Python	File Handling	➢ read/write files, context manager
Programming	Python	Exception Handling	➢ try, except, else, finally
Programming	Python	Custom Exceptions	➢ raising user-defined errors
Programming	Python	OOP Basics	➢ class, object, __init__
Programming	Python	OOP Advanced	➢ inheritance, polymorphism

Category	Library / Tool	Topic	➢ Sub-Topic / Depth (What you MUST know)
Programming	Python	Encapsulation	➢ public, protected, private
Programming	Python	Abstraction	➢ abstract classes, interfaces
Programming	Python	Magic Methods	➢ __str__, __repr__, __len__
Programming	Python	Memory Model	➢ stack vs heap, references
Programming	Python	Virtual Environment	➢ venv, conda, environment isolation
Programming	Python	Package Management	➢ pip, requirements.txt
Programming	Python	Debugging	➢ traceback, logging basics
Programming	Python	Performance Basics	➢ time vs space complexity intuition
Programming	Git	Version Control Basics	➢ init, clone, repository structure
Programming	Git	Staging	➢ add, .gitignore
Programming	Git	Commits	➢ commit, commit messages
Programming	Git	Branching	➢ create, switch, merge
Programming	Git	Merge Conflicts	➢ resolving conflicts
Programming	Git	Rebase	➢ linear history
Programming	GitHub	Remote Repos	➢ push, pull, fetch
Programming	GitHub	Workflow	➢ feature branches
Programming	GitHub	Pull Requests	➢ code review process
Programming	GitHub	Issues	➢ bug tracking
Programming	GitHub	Collaboration	➢ team workflows

□ MATHEMATICS FOR MACHINE LEARNING — ENLARGED

Category	**Sub-Category**	**Topic**	➢ **Depth / What You MUST Understand**
Linear Algebra	Scalars & Vectors	Scalars vs vectors	➢ Physical & geometric meaning
Linear Algebra	Vectors	Vector addition	➢ Direction & magnitude
Linear Algebra	Vectors	Dot product	➢ Angle, similarity, projection
Linear Algebra	Vectors	Norms	➢ L1, L2 norms, distance meaning
Linear Algebra	Vectors	Cosine similarity	➢ Used in NLP, embeddings
Linear Algebra	Matrices	Matrix representation	➢ Data as matrices
Linear Algebra	Matrices	Matrix multiplication	➢ Why order matters
Linear Algebra	Matrices	Transpose	➢ Shape transformation
Linear Algebra	Matrices	Identity matrix	➢ Neutral element
Linear Algebra	Matrices	Inverse matrix	➢ Solving linear equations
Linear Algebra	Matrices	Rank of matrix	➢ Feature redundancy
Linear Algebra	Eigen Concepts	Eigenvalues	➢ Variance direction

Category	Sub-Category	Topic	➢ **Depth / What You MUST Understand**
Linear Algebra	Eigen Concepts	Eigenvectors	➢ Principal directions
Linear Algebra	PCA	Dimensionality reduction	➢ Variance maximization
Linear Algebra	PCA	Covariance matrix	➢ Feature correlation
Linear Algebra	PCA	Explained variance	➢ Feature selection
Probability	Random Variables	Discrete vs continuous	➢ Modeling uncertainty
Probability	Distributions	Bernoulli	➢ Binary outcomes
Probability	Distributions	Binomial	➢ Repeated trials
Probability	Distributions	Gaussian (Normal)	➢ Noise modeling
Probability	Distributions	Poisson	➢ Event occurrence
Probability	Probability Rules	Joint probability	➢ Combined events
Probability	Probability Rules	Conditional probability	➢ Given information
Probability	Probability Rules	Bayes theorem	➢ Posterior inference
Probability	Probability Rules	Independence	➢ Naive Bayes assumption
Statistics	Central Tendency	Mean, median, mode	➢ Distribution center
Statistics	Dispersion	Variance	➢ Spread of data
Statistics	Dispersion	Standard deviation	➢ Uncertainty
Statistics	Sampling	Sampling methods	➢ Bias awareness
Statistics	Estimation	Point estimation	➢ Single value estimate
Statistics	Estimation	Confidence interval	➢ Uncertainty range

Category	**Sub-Category**	**Topic**	➢ **Depth / What You MUST Understand**
Statistics	Hypothesis Testing	Null & alternative	➢ Statistical decisions
Statistics	Hypothesis Testing	p-value	➢ Significance
Statistics	Hypothesis Testing	Type I & II errors	➢ False decisions
Statistics	Model Evaluation	Correlation	➢ Linear relationship
Statistics	Model Evaluation	Covariance	➢ Feature dependency
Optimization	Functions	Objective function	➢ What ML minimizes
Optimization	Loss Functions	MSE	➢ Regression loss
Optimization	Loss Functions	Log loss	➢ Classification loss
Optimization	Loss Functions	Cross-entropy	➢ Probabilistic models
Optimization	Gradients	Partial derivatives	➢ Direction of steepest descent
Optimization	Algorithms	Gradient descent	➢ Core ML optimizer
Optimization	Algorithms	Batch GD	➢ Stability
Optimization	Algorithms	Stochastic GD	➢ Speed vs noise
Optimization	Algorithms	Mini-batch GD	➢ Practical training
Optimization	Learning Rate	Learning rate tuning	➢ Convergence vs divergence
Optimization	Optimization Issues	Local minima	➢ Non-convex losses
Optimization	Optimization Issues	Saddle points	➢ Flat regions

Category	Sub-Category	Topic	➢ Depth / What You MUST Understand
Optimization	Regularization	L1 regularization	➢ Feature sparsity
Optimization	Regularization	L2 regularization	➢ Weight decay
Optimization	Bias–Variance	Underfitting	➢ High bias
Optimization	Bias–Variance	Overfitting	➢ High variance
Optimization	Bias–Variance	Trade-off	➢ Generalization control
Information Theory	Entropy	Entropy	➢ Measure of uncertainty
Information Theory	Entropy	Information gain	➢ Decision trees
Information Theory	KL Divergence	Distribution difference	➢ Model comparison

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🤘 **CORE MACHINE LEARNING (sklearn) — ENLARGED**

Category	Algorithm / Concept	Topic	➢ Depth / What You MUST Understand
ML Foundations	Problem Types	Supervised learning	➢ Labeled data, prediction tasks
ML Foundations	Problem Types	Unsupervised learning	➢ Pattern discovery

Category	Algorithm / Concept	Topic	➢ Depth / What You MUST Understand
ML Foundations	Problem Types	Semi-supervised learning	➢ Limited labels
ML Foundations	Problem Types	Classification vs Regression	➢ Output types
ML Foundations	ML Pipeline	Train–test split	➢ Prevent data leakage
ML Foundations	ML Pipeline	Cross-validation	➢ Reliable performance
ML Foundations	ML Pipeline	Hyperparameter tuning	➢ Bias–variance control
Regression	Linear Regression	Model formulation	➢ Line fitting intuition
Regression	Linear Regression	Assumptions	➢ Linearity, independence
Regression	Linear Regression	Cost function	➢ MSE
Regression	Linear Regression	Multicollinearity	➢ Feature correlation
Regression	Ridge Regression	L2 regularization	➢ Shrink weights
Regression	Lasso Regression	L1 regularization	➢ Feature selection
Regression	Elastic Net	L1 + L2	➢ Balanced regularization
Classification	Logistic Regression	Sigmoid function	➢ Probability output
Classification	Logistic Regression	Decision boundary	➢ Linear separation
Classification	Logistic Regression	Log loss	➢ Why not MSE

Category	**Algorithm / Concept**	**Topic**	➢ **Depth / What You MUST Understand**
Classification	KNN	Distance metrics	➢ Euclidean, Manhattan
Classification	KNN	K selection	➢ Bias vs variance
Classification	KNN	Curse of dimensionality	➢ High-dim data
Probabilistic	Naive Bayes	Bayes theorem	➢ Posterior probability
Probabilistic	Naive Bayes	Conditional independence	➢ Core assumption
Probabilistic	Gaussian NB	Continuous features	➢ Distribution modeling
Margin Methods	SVM	Maximum margin	➢ Robust separation
Margin Methods	SVM	Kernel trick	➢ Non-linear boundaries
Margin Methods	SVM	Soft margin	➢ Handling noise
Tree Models	Decision Tree	Tree structure	➢ Nodes, leaves
Tree Models	Decision Tree	Split criteria	➢ Gini vs entropy
Tree Models	Decision Tree	Overfitting	➢ Depth control
Ensemble	Bagging	Bootstrap sampling	➢ Variance reduction
Ensemble	Random Forest	Feature randomness	➢ Decorrelated trees
Ensemble	Random Forest	OOB error	➢ Internal validation
Ensemble	Random Forest	Feature importance	➢ Interpretability
Boosting	Gradient Boosting	Sequential learning	➢ Error correction
Boosting	AdaBoost	Weight updates	➢ Focus on hard samples
Boosting	XGBoost	Regularization	➢ Prevent overfitting
Boosting	XGBoost	Tree pruning	➢ Model efficiency

Category	**Algorithm / Concept**	**Topic**	➢ **Depth / What You MUST Understand**
Boosting	LightGBM	Leaf-wise growth	➢ Speed vs overfit
Dimensionality Reduction	PCA	Variance maximization	➢ Feature compression
Dimensionality Reduction	PCA	Eigen decomposition	➢ Math link
Dimensionality Reduction	PCA	When NOT to use	➢ Interpretability loss
Clustering	K-Means	Centroid update	➢ Iterative convergence
Clustering	K-Means	Choosing K	➢ Elbow method
Clustering	K-Means	Limitations	➢ Non-spherical clusters
Clustering	Hierarchical	Agglomerative	➢ Dendrogram
Clustering	DBSCAN	Density based	➢ Noise handling
Model Evaluation	Metrics	Accuracy	➢ Balanced data
Model Evaluation	Metrics	Precision & Recall	➢ Imbalanced data
Model Evaluation	Metrics	F1 score	➢ Tradeoff
Model Evaluation	Metrics	ROC-AUC	➢ Ranking quality
Model Evaluation	Metrics	Confusion matrix	➢ Error analysis
Feature Engineering	Scaling	Standardization	➢ Distance-based models
Feature Engineering	Scaling	Normalization	➢ Range control
Feature Engineering	Encoding	One-hot encoding	➢ Categorical data
Feature Engineering	Encoding	Label encoding	➢ Ordinal data

Category	Algorithm / Concept	Topic	➢ Depth / What You MUST Understand
Feature Engineering	Feature Selection	Filter methods	➢ Statistical tests
Feature Engineering	Feature Selection	Wrapper methods	➢ Model-based
Feature Engineering	Feature Selection	Embedded methods	➢ Lasso, trees
ML Pitfalls	Data Leakage	Leakage sources	➢ False accuracy
ML Pitfalls	Overfitting	Detection	➢ Train vs test gap
ML Pitfalls	Underfitting	Detection	➢ Poor training perf
ML Pitfalls	Imbalanced Data	Handling	➢ Class weights
ML Pitfalls	Bias	Sampling bias	➢ Dataset issues

📊 DATA ENGINEERING FOR ML — ENLARGED

Category	Tool / Technology	Topic	➢ Depth / What You MUST Understand
Data Fundamentals	Data Types	Structured data	➢ Tables, rows, columns
Data Fundamentals	Data Types	Semi-structured data	➢ JSON, logs
Data Fundamentals	Data Types	Unstructured data	➢ Text, images
Data Collection	Data Sources	CSV, Excel	➢ Ingestion basics

Category	Tool / Technology	Topic	➢ Depth / What You MUST Understand
Data Collection	APIs	REST APIs	➢ Pagination, rate limits
Data Collection	Web Scraping	Data extraction	➢ Ethics & legality
Data Ingestion	Batch vs Streaming	Batch ingestion	➢ Offline training
Data Ingestion	Batch vs Streaming	Streaming ingestion	➢ Real-time ML
Python Data	NumPy	ndarrays	➢ Vectorized operations
Python Data	NumPy	Broadcasting	➢ Shape compatibility
Python Data	NumPy	Performance	➢ Loop vs vector
Python Data	Pandas	DataFrames	➢ Indexing, slicing
Python Data	Pandas	Data cleaning	➢ Missing values
Python Data	Pandas	Data transformation	➢ groupby, apply
Python Data	Pandas	Feature engineering	➢ Derived features
Python Data	Pandas	Time series data	➢ Datetime ops
Data Quality	Data Validation	Schema checks	➢ Data consistency
Data Quality	Outliers	Detection	➢ IQR, z-score
Data Quality	Noise	Handling	➢ Smoothing
Databases	SQL	SELECT queries	➢ Filtering
Databases	SQL	JOINs	➢ Inner, outer
Databases	SQL	Subqueries	➢ Nested logic
Databases	SQL	Indexing	➢ Performance tuning

Category	**Tool / Technology**	**Topic**	➢ **Depth / What You MUST Understand**
Databases	SQL	Window functions	➢ Analytics
Data Modeling	Relational Models	Normalization	➢ Reduce redundancy
Data Modeling	Star Schema	Analytics use	➢ Fact & dimension
Feature Stores	Feature Engineering	Reusability	➢ Training vs inference
Feature Stores	Feature Versioning	Consistency	➢ Reproducibility
Pipelines	ETL	Extract	➢ Source systems
Pipelines	ETL	Transform	➢ Cleaning, encoding
Pipelines	ETL	Load	➢ Data warehouse
Pipelines	ELT	Modern pipelines	➢ Cloud-based
Workflow Orchestration	Airflow	DAGs	➢ Scheduling
Workflow Orchestration	Cron	Job scheduling	➢ Simpler flows
Big Data	Distributed Systems	Scale-out	➢ Large datasets
Big Data	Spark	DataFrames	➢ Parallel compute
Big Data	Spark	ML integration	➢ Large-scale ML
Data Storage	Data Lakes	Raw data	➢ Scalability
Data Storage	Data Warehouses	Structured data	➢ Analytics
Data Versioning	DVC	Dataset versioning	➢ Reproducibility

Category	Tool / Technology	Topic	➢ Depth / What You MUST Understand
Data Security	Access Control	Permissions	➢ Data privacy
Data Governance	Compliance	GDPR basics	➢ Responsible data
ML Data	Train/Test Split	Temporal split	➢ Time leakage
ML Data	Data Leakage	Leakage sources	➢ Label leakage
ML Data	Sampling	Stratified sampling	➢ Imbalanced data
ML Data	Feature Scaling	When to scale	➢ Model dependency
ML Data	Categorical Data	Encoding strategies	➢ One-hot vs target
Production Data	Online vs Offline Data	Serving mismatch	➢ Training vs inference
Production Data	Data Drift	Detection	➢ Distribution shift
Production Data	Data Monitoring	Quality metrics	➢ Production checks

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□ DEEP LEARNING (PyTorch) — ENLARGED

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
DL Foundations	Deep Learning	Why DL	➢ Non-linear representation power
DL Foundations	Neural Networks	Perceptron	➢ Single neuron intuition
DL Foundations	Neural Networks	Multi-Layer Networks	➢ Universal approximation idea
DL Foundations	PyTorch	Tensors	➢ Shape, dtype, device (CPU/GPU)
DL Foundations	PyTorch	Tensor operations	➢ Broadcasting, slicing
Autograd	PyTorch	Computational graph	➢ Dynamic graph concept
Autograd	PyTorch	Automatic differentiation	➢ requires_grad
Autograd	PyTorch	Gradient flow	➢ How gradients propagate
Training	DL Theory	Forward pass	➢ Data → prediction
Training	DL Theory	Backpropagation	➢ Chain rule intuition
Training	DL Theory	Gradient descent	➢ Parameter updates
Activations	DL Theory	Sigmoid	➢ Saturation issue
Activations	DL Theory	Tanh	➢ Centered output
Activations	DL Theory	ReLU	➢ Vanishing gradient fix
Activations	DL Theory	Leaky ReLU	➢ Dead neuron issue
Loss Functions	DL Theory	MSE	➢ Regression use
Loss Functions	DL Theory	Cross-entropy	➢ Classification
Loss Functions	DL Theory	Negative log likelihood	➢ Probabilistic view

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Optimization	PyTorch	SGD	➢ Stability vs speed
Optimization	PyTorch	Momentum	➢ Faster convergence
Optimization	PyTorch	Adam	➢ Adaptive learning rates
Learning Rate	DL Theory	LR scheduling	➢ Convergence control
Initialization	DL Theory	Weight initialization	➢ Xavier, He
Regularization	DL Theory	L1/L2 regularization	➢ Weight control
Regularization	DL Theory	Dropout	➢ Prevent overfitting
Regularization	DL Theory	Batch Normalization	➢ Training stability
Model Building	PyTorch	nn.Module	➢ Custom models
Model Building	PyTorch	Layers	➢ Linear, Conv
Model Building	PyTorch	Sequential vs subclassing	➢ Flexibility
Training Loop	PyTorch	Custom training loop	➢ Full control
Training Loop	PyTorch	Zeroing gradients	➢ Gradient accumulation
Training Loop	PyTorch	Backward pass	➢ loss.backward()
Training Loop	PyTorch	Optimizer step	➢ optimizer.step()
Evaluation	PyTorch	model.eval()	➢ Inference mode
Evaluation	PyTorch	torch.no_grad()	➢ Memory efficiency
Overfitting	DL Theory	Detection	➢ Train vs val loss
Underfitting	DL Theory	Detection	➢ Poor learning
Debugging	DL Practice	Exploding gradients	➢ Gradient clipping
Debugging	DL Practice	Vanishing gradients	➢ Activation choice
Hardware	PyTorch	GPU usage	➢ CUDA basics

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Hardware	PyTorch	Device transfer	➢ .to(device)
Model Saving	PyTorch	Saving models	➢ state_dict
Model Loading	PyTorch	Loading models	➢ Reproducibility
Reproducibility	DL Practice	Random seeds	➢ Deterministic runs
Performance	DL Practice	Batch size	➢ Memory vs speed
Performance	DL Practice	Mixed precision	➢ Speed-ups

👁 COMPUTER VISION — ENLARGED

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
CV Foundations	CV Theory	What is an image	➢ Pixels, channels, resolution
CV Foundations	CV Theory	Color spaces	➢ RGB, Grayscale, HSV
CV Foundations	CV Theory	Image representation	➢ Matrix form
CV Foundations	OpenCV	Image I/O	➢ Read, write, display
CV Foundations	OpenCV	Image resizing	➢ Interpolation methods
CV Foundations	OpenCV	Image normalization	➢ Pixel scaling
Classical CV	OpenCV	Edge detection	➢ Sobel, Canny
Classical CV	OpenCV	Blurring	➢ Gaussian, median
Classical CV	OpenCV	Thresholding	➢ Binary segmentation

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Classical CV	OpenCV	Morphology	➢ Erosion, dilation
Classical CV	OpenCV	Contours	➢ Shape extraction
CNN Basics	DL Theory	Convolution	➢ Filters, stride, padding
CNN Basics	DL Theory	Feature maps	➢ Hierarchical features
CNN Basics	DL Theory	Receptive field	➢ Context growth
CNN Basics	DL Theory	Pooling	➢ Max vs average
CNN Basics	DL Theory	Translation invariance	➢ Spatial robustness
CNN Training	PyTorch	CNN implementation	➢ From scratch
CNN Training	PyTorch	Loss for images	➢ Cross-entropy
CNN Training	PyTorch	Optimizers	➢ Adam, SGD
CNN Training	PyTorch	Overfitting in CNNs	➢ Detection
Transfer Learning	torchvision	Pretrained models	➢ Why it works
Transfer Learning	torchvision	Fine-tuning	➢ Freeze vs train
Transfer Learning	torchvision	Feature extraction	➢ Shallow vs deep
CNN Architectures	CV Models	LeNet	➢ Early CNN
CNN Architectures	CV Models	AlexNet	➢ Deep CNN
CNN Architectures	CV Models	VGG	➢ Deep stacks
CNN Architectures	CV Models	ResNet	➢ Skip connections
CNN Architectures	CV Models	EfficientNet	➢ Scaling rules

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Object Detection	CV Theory	Detection vs classification	➢ Localization
Object Detection	CV Theory	Bounding boxes	➢ Coordinates
Object Detection	YOLO	One-stage detection	➢ Speed advantage
Object Detection	YOLO	Anchor boxes	➢ Size priors
Object Detection	YOLO	IoU	➢ Overlap metric
Object Detection	YOLO	NMS	➢ Duplicate removal
Object Detection	CV Metrics	mAP	➢ Detection accuracy
Image Segmentation	CV Theory	Semantic segmentation	➢ Pixel labels
Image Segmentation	CV Theory	Instance segmentation	➢ Object masks
Image Segmentation	CV Models	U-Net	➢ Medical imaging
Image Segmentation	CV Models	Mask R-CNN	➢ Instance masks
Data Handling	CV Practice	Dataset annotation	➢ Label quality
Data Handling	CV Practice	Data augmentation	➢ Flip, rotate
Data Handling	CV Practice	Class imbalance	➢ Sampling
Model Evaluation	CV Metrics	Accuracy	➢ Balanced sets
Model Evaluation	CV Metrics	Precision / Recall	➢ Detection
Model Evaluation	CV Metrics	Confusion matrix	➢ Error analysis
Deployment CV	CV Practice	Image preprocessing	➢ Production parity
Deployment CV	CV Practice	Batch vs real-time	➢ Latency tradeoff

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Debugging CV	CV Practice	Misclassification	➢ Error analysis
Debugging CV	CV Practice	Overfitting	➢ Visual diagnosis

📝 NLP & TRANSFORMERS — ENLARGED

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
NLP Foundations	NLP Theory	What is NLP	➢ Language as data
NLP Foundations	NLP Theory	Text preprocessing	➢ Lowercasing, cleaning
NLP Foundations	NLP Theory	Tokenization	➢ Word vs subword
NLP Foundations	NLP Theory	Stopwords	➢ When to remove
NLP Foundations	NLP Theory	Stemming	➢ Rule-based
NLP Foundations	NLP Theory	Lemmatization	➢ Meaning-preserving
NLP Foundations	NLP Theory	Vocabulary	➢ OOV problem
Classical NLP	NLP Theory	Bag of Words	➢ Sparse vectors
Classical NLP	NLP Theory	TF-IDF	➢ Importance weighting

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Classical NLP	NLP Theory	N-grams	➢ Context window
Classical NLP	NLP Theory	Limitations	➢ Loss of semantics
Embeddings	NLP Theory	Word embeddings	➢ Dense representations
Embeddings	Gensim	Word2Vec	➢ CBOW vs Skip-gram
Embeddings	Gensim	GloVe	➢ Global co-occurrence
Embeddings	NLP Theory	FastText	➢ Subword info
Embeddings	NLP Theory	Semantic similarity	➢ Cosine similarity
Sequence Models	NLP Theory	Sequence modeling	➢ Order matters
Sequence Models	NLP Theory	RNN	➢ Vanishing gradient
Sequence Models	NLP Theory	LSTM	➢ Long-term memory
Sequence Models	NLP Theory	GRU	➢ Simplified LSTM
Attention	NLP Theory	Attention mechanism	➢ Focus on relevant tokens
Attention	NLP Theory	Self-attention	➢ Intra-sentence relations
Transformers	NLP Theory	Transformer architecture	➢ Encoder–decoder
Transformers	NLP Theory	Positional encoding	➢ Order information
Transformers	NLP Theory	Multi-head attention	➢ Parallel focus
Transformers	NLP Theory	Feed-forward layers	➢ Token-wise transform
Transformers	NLP Theory	Layer normalization	➢ Training stability

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Transformers	NLP Theory	Why transformers scale	➢ Parallelism
Pretrained Models	Hugging Face	BERT	➢ Bidirectional encoding
Pretrained Models	Hugging Face	RoBERTa	➢ Improved BERT
Pretrained Models	Hugging Face	DistilBERT	➢ Lightweight models
Pretrained Models	Hugging Face	GPT (conceptual)	➢ Autoregressive
Fine-Tuning	Transformers	Transfer learning	➢ Task-specific heads
Fine-Tuning	Transformers	Token classification	➢ NER
Fine-Tuning	Transformers	Sequence classification	➢ Sentiment
Fine-Tuning	Transformers	Question answering	➢ Span prediction
Training NLP	Transformers	Loss functions	➢ Cross-entropy
Training NLP	Transformers	Padding & masking	➢ Batch handling
Training NLP	Transformers	Gradient accumulation	➢ Memory control
Evaluation	NLP Metrics	Accuracy	➢ Balanced tasks
Evaluation	NLP Metrics	Precision / Recall	➢ NER
Evaluation	NLP Metrics	F1 score	➢ Imbalanced classes
Evaluation	NLP Metrics	BLEU	➢ Text generation
Evaluation	NLP Metrics	ROUGE	➢ Summarization

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Deployment NLP	NLP Practice	Tokenizer consistency	➢ Train vs inference
Deployment NLP	NLP Practice	Max sequence length	➢ Truncation
Deployment NLP	NLP Practice	Latency	➢ Model size impact
NLP Failures	NLP Practice	Hallucination	➢ LLM risk
NLP Failures	NLP Practice	Bias in text	➢ Dataset issues

□ GENERATIVE AI & RAG — ENLARGED

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
GenAI Foundations	GenAI Theory	What is Generative AI	➢ Data → distribution → generation
GenAI Foundations	GenAI Theory	Discriminative vs Generative	➢ Predict vs generate
GenAI Foundations	GenAI Theory	Language modeling	➢ Next-token prediction
GenAI Foundations	GenAI Theory	Autoregressive models	➢ Causal decoding

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
GenAI Foundations	GenAI Theory	Sampling methods	➢ Greedy, top-k, top-p
GenAI Foundations	GenAI Theory	Temperature	➢ Creativity vs accuracy
LLM Architecture	LLM Theory	Transformer decoder	➢ GPT-style models
LLM Architecture	LLM Theory	Attention scaling	➢ Context length
LLM Architecture	LLM Theory	Tokenization	➢ BPE, WordPiece
LLM Architecture	LLM Theory	Context window	➢ Limits & truncation
LLM Architecture	LLM Theory	KV cache	➢ Inference speedup
Prompt Engineering	Prompting	Prompt structure	➢ System, user, context
Prompt Engineering	Prompting	Zero-shot prompting	➢ No examples
Prompt Engineering	Prompting	Few-shot prompting	➢ In-context learning
Prompt Engineering	Prompting	Chain-of-thought	➢ Reasoning guidance
Prompt Engineering	Prompting	Prompt injection	➢ Security risks
Embeddings	SentenceTransformers	Text embeddings	➢ Semantic vectors
Embeddings	SentenceTransformers	Similarity search	➢ Cosine distance
Embeddings	GenAI Theory	Embedding drift	➢ Model updates

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Vector Databases	FAISS	Vector indexing	➢ Flat vs IVF
Vector Databases	FAISS	ANN search	➢ Speed vs accuracy
Vector Databases	Chroma	Metadata filtering	➢ Hybrid search
Vector Databases	Vector DB Theory	Vector normalization	➢ Distance stability
RAG Foundations	RAG Theory	What is RAG	➢ Retrieval + generation
RAG Foundations	RAG Theory	Why RAG is needed	➢ Hallucination control
RAG Ingestion	RAG Pipeline	Document loaders	➢ PDF, HTML, DOC
RAG Ingestion	RAG Pipeline	Text chunking	➢ Chunk size & overlap
RAG Ingestion	RAG Pipeline	Metadata design	➢ Source attribution
Retrieval	RAG Pipeline	Dense retrieval	➢ Embedding search
Retrieval	RAG Pipeline	Top-k retrieval	➢ Recall vs latency
Retrieval	RAG Pipeline	Hybrid retrieval	➢ Keyword + vector
Context Building	RAG Pipeline	Context window packing	➢ Token budgeting
Context Building	RAG Pipeline	Reranking	➢ Improve relevance
Generation	RAG Pipeline	Prompt + context	➢ Controlled output
Evaluation	RAG Eval	Faithfulness	➢ Grounded answers
Evaluation	RAG Eval	Relevance	➢ Answer quality

Category	Tool / Library	Topic	➢ Depth / What You MUST Understand
Evaluation	RAG Eval	Context recall	➢ Retrieval quality
Optimization	RAG Optimization	Chunk tuning	➢ Precision tradeoff
Optimization	RAG Optimization	Retrieval tuning	➢ k, distance metric
Optimization	RAG Optimization	Caching	➢ Latency reduction
Production RAG	GenAI Ops	Latency control	➢ Streaming
Production RAG	GenAI Ops	Cost optimization	➢ Token usage
Production RAG	GenAI Ops	Fallback strategies	➢ No-answer cases
Security	GenAI Safety	Data leakage	➢ Private data safety
Security	GenAI Safety	Prompt injection defense	➢ Guardrails
Security	GenAI Safety	PII filtering	➢ Compliance
Advanced GenAI	GenAI Theory	Tool calling	➢ Function calling
Advanced GenAI	GenAI Theory	Agents	➢ Multi-step reasoning
Advanced GenAI	GenAI Theory	Memory	➢ Conversation state

🎮 **REINFORCEMENT LEARNING — ENLARGED**

Category	Library / Theory	Topic	➢ Depth / What You MUST Understand
RL Foundations	RL Theory	What is RL	➢ Learning via interaction
RL Foundations	RL Theory	Agent–Environment	➢ Observation → action → reward
RL Foundations	RL Theory	Reward signal	➢ Sparse vs dense rewards
RL Foundations	RL Theory	Episodes	➢ Terminal states
RL Theory	RL Theory	Markov Property	➢ Memoryless assumption
RL Theory	RL Theory	MDP	➢ States, actions, rewards, transitions
RL Theory	RL Theory	Discount factor (γ)	➢ Future reward importance
RL Theory	RL Theory	Policy	➢ Deterministic vs stochastic
RL Theory	RL Theory	Value function	➢ State-value
RL Theory	RL Theory	Action-value	➢ Q-function
Exploration	RL Theory	Exploration vs exploitation	➢ Tradeoff
Exploration	RL Theory	ε-greedy	➢ Random action strategy
Exploration	RL Theory	Softmax exploration	➢ Probability-based
Value-Based	Q-Learning	Q-table	➢ State–action storage
Value-Based	Q-Learning	Bellman equation	➢ Recursive update
Value-Based	Q-Learning	Off-policy learning	➢ Learn optimal policy
Value-Based	SARSA	On-policy learning	➢ Follow current policy
Value-Based	SARSA	Q update difference	➢ SARSA vs Q-learning

Category	Library / Theory	Topic	➢ Depth / What You MUST Understand
Deep RL	Deep RL	Function approximation	➢ Replace Q-table
Deep RL	DQN	Neural networks	➢ Approximate Q-values
Deep RL	DQN	Experience replay	➢ Break correlations
Deep RL	DQN	Target network	➢ Training stability
Policy-Based	Policy Gradient	REINFORCE	➢ Monte Carlo gradients
Policy-Based	Policy Gradient	Variance reduction	➢ Baseline
Actor-Critic	RL Theory	Actor–Critic	➢ Value + policy
Advanced RL	PPO	Clipped objective	➢ Stability
Environments	OpenAI Gym	Built-in envs	➢ CartPole, LunarLander
Environments	OpenAI Gym	Custom envs	➢ Design states & rewards
RL Failures	RL Practice	Reward hacking	➢ Bad incentives
RL Failures	RL Practice	Non-stationarity	➢ Changing dynamics

🌐 **DEPLOYMENT & BACKEND — ENLARGED**

Category	Library / Tool	Topic	➢ Depth / What You MUST Understand
Backend Basics	Backend Theory	Why deployment	➢ Model → product
APIs	Flask	REST API	➢ Request–response

Category	Library / Tool	Topic	➢ Depth / What You MUST Understand
APIs	Flask	Model inference	➢ Loading & predicting
APIs	Flask	Error handling	➢ Invalid inputs
APIs	FastAPI	Async APIs	➢ High concurrency
APIs	FastAPI	Data validation	➢ Pydantic models
REST	HTTP	HTTP methods	➢ GET, POST, PUT, DELETE
REST	HTTP	Status codes	➢ 200, 400, 500
REST	REST Design	Idempotency	➢ Safe APIs
Serialization	JSON	Input format	➢ Schema design
Serialization	JSON	Output format	➢ Prediction response
Performance	Backend	Latency	➢ Inference time
Performance	Backend	Throughput	➢ Requests/sec
Performance	Backend	Batch inference	➢ Efficiency
Performance	Backend	Caching	➢ Faster responses
Security	Backend	Input sanitization	➢ Injection risks
Security	Backend	Auth basics	➢ API keys
Production	Backend	Logging	➢ Debugging
Production	Backend	Health checks	➢ Service monitoring

🐳 DOCKER, LINUX & CLOUD — ENLARGED

Category	Tool	Topic	➢ Depth / What You MUST Understand
Containers	Docker	What is Docker	➢ Container vs VM
Containers	Docker	Dockerfile	➢ Image instructions
Containers	Docker	Base images	➢ Python slim
Containers	Docker	Image layers	➢ Caching
Containers	Docker	Image optimization	➢ Size reduction
Containers	Docker	Multi-stage builds	➢ Clean images
OS	Linux	File system	➢ Paths & permissions
OS	Linux	Core commands	➢ ls, cd, cp, rm
OS	Linux	Process management	➢ ps, top, kill
OS	Linux	Networking	➢ Ports
OS	Linux	Logs	➢ Debugging
Cloud	Cloud Basics	Cloud computing	➢ IaaS, PaaS
Cloud	AWS/GCP	Compute	➢ EC2, VM
Cloud	AWS/GCP	Containers	➢ ECS, GKE
Cloud	Cloud Storage	Object storage	➢ S3, GCS
Cloud	Cloud Networking	Load balancer	➢ Traffic routing
Cloud	Cloud Security	IAM	➢ Permissions
Cloud	Deployment	CI/CD to cloud	➢ Automation

🔁 MLOps — ENLARGED

Category	Library / Tool	Topic	➢ Depth / What You MUST Understand
MLOps Foundations	MLOps Theory	Why MLOps	➢ ML in production
Experimentation	MLflow	Experiment tracking	➢ Params, metrics
Experimentation	MLflow	Model registry	➢ Versioning
Versioning	DVC	Data versioning	➢ Reproducibility
Versioning	DVC	Model versioning	➢ Rollback
Pipelines	CI/CD	Training pipelines	➢ Automation
Pipelines	CI/CD	Testing	➢ Model validation
Monitoring	MLOps	Model drift	➢ Concept drift
Monitoring	MLOps	Data drift	➢ Distribution shift
Monitoring	MLOps	Performance decay	➢ Accuracy drop
Retraining	MLOps	Scheduled retraining	➢ Fresh data
Retraining	MLOps	Trigger-based retraining	➢ Drift detection
Governance	MLOps	Model lineage	➢ Traceability
Governance	MLOps	Auditability	➢ Compliance
Production	MLOps	Canary deployment	➢ Safe rollout
Production	MLOps	Shadow deployment	➢ Compare models

☎️ **ML SYSTEM DESIGN — ENLARGED**

Category	Tool / Concept	Topic	Depth / What You MUST Understand
System Basics	Architecture	End-to-end ML	Data → model → API
System Basics	Architecture	Offline pipeline	Training flow
System Basics	Architecture	Online pipeline	Inference flow
Inference	System Design	Batch inference	Cost efficiency
Inference	System Design	Real-time inference	Low latency
Scaling	System Design	Horizontal scaling	Multiple replicas
Scaling	System Design	Load balancing	Traffic distribution
Scaling	System Design	Auto-scaling	Demand spikes
Reliability	System Design	Fault tolerance	Failures
Safety	System Design	Model rollback	Bad model recovery
Safety	System Design	A/B testing	Model comparison
Monitoring	System Design	Metrics	Latency, accuracy
Monitoring	System Design	Logging	Debugging
Monitoring	System Design	Alerting	Production issues
Cost	System Design	Cost optimization	Cloud spend
Security	System Design	Data privacy	PII protection